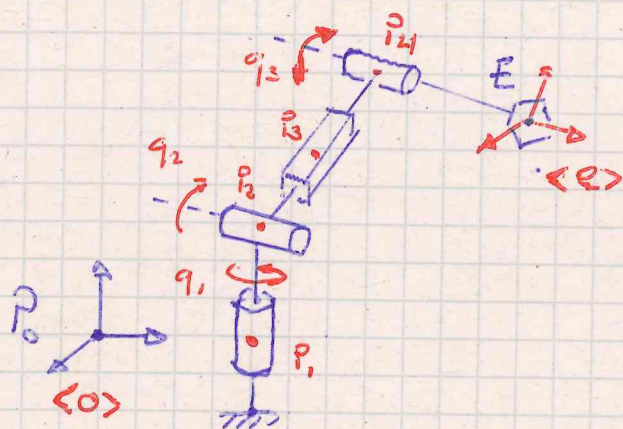


THE JACOBIAN OPERATOR

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Consider $\underline{z}_{i/0} = (P_i - P_0)$
for $i = 1 \div n$

Let's compute $\frac{d}{dt} \underline{z}_{i/0} = \underline{v}_{i/0}$
and $\underline{\omega}_{i/0}$

$$\begin{cases} \underline{v}_{i/0} = \frac{d_0}{dt} \underline{z}_{i/0} = \frac{d_0}{dt} \underline{z}_{1/0} + \frac{d_0}{dt} \underline{z}_{i/1} = \sum_{j=1}^i \frac{d_j}{dt} \underline{z}_{j/j-1} + \sum_{j=1}^i \underline{\omega}_{j/0} \times \underline{z}_{j/j-1} \\ \underline{\omega}_{i/0} = \sum_{j=1}^i \underline{\omega}_{j/j-1} \end{cases}$$

$$\uparrow \frac{d_1}{dt} \underline{z}_{1/0} + \underline{\omega}_{1/0} \times \underline{z}_{1/0}$$

$$\uparrow \frac{d_2}{dt} \underline{z}_{2/1} + \frac{d_2}{dt} \underline{z}_{i/2}$$

$$\uparrow \frac{d_2}{dt} \underline{z}_{1/2} + \underline{\omega}_{2/2} \times \underline{z}_{1/2}$$

$$\uparrow \frac{d_2}{dt} \underline{z}_{3/2} + \frac{d_2}{dt} \underline{z}_{i/3}$$

Notice that :

$$\frac{d_j}{dt} \underline{z}_{j/j-1} = \begin{cases} 0 & j \in R \\ K_j \dot{q}_j & j \in T \end{cases}$$

$$\underline{\omega}_{j/j-1} = \begin{cases} 0 & j \in T \\ K_j \dot{q}_j & j \in R \end{cases}$$

$$\underline{v}_{i/0} = \sum_{j=1}^i \frac{d_j}{dt} \underline{z}_{j/j-1} + \sum_{j=1}^i \underline{w}_{j/j-1} \times \underline{z}_{i/j}$$

$$\begin{array}{cc} \underline{0} \text{ if } i \notin \mathcal{N} & \underline{K_j' \dot{q}_j} \text{ if } i \in \mathcal{N} \\ \underline{K_j' \dot{q}_j} \text{ if } i \in \mathcal{N} & \underline{0} \text{ if } i \notin \mathcal{N} \end{array}$$

$$\underline{w}_{i/0} = \sum_{j=1}^i \underline{w}_{j/j-1}$$

Then we can define the operators

$$\underline{J}_{i/j}^L = \begin{cases} \underline{K_j} & j \in \mathcal{T} \\ (\underline{K_j} \times \underline{z}_{i/j}) & j \in \mathcal{N} \end{cases}$$

$$\underline{J}_{i/j}^L = \underline{0} \quad j > i$$

$$\underline{J}_{i/j}^A = \begin{cases} \underline{0} & j \in \mathcal{T} \\ \underline{K_j} & j \in \mathcal{N} \end{cases}$$

$$\underline{J}_{i/j}^A = \underline{0} \quad j > i$$

Then:

$$\underline{v}_{i/0} = \sum_{j=1}^i \underline{J}_{i/j-1}^L \dot{q}_j \Rightarrow \underline{v}_{i/0} = \left[\underline{J}_{i/1}^L \quad \underline{J}_{i/2}^L \quad \dots \quad \underline{J}_{i/i}^L \right] \begin{bmatrix} \dot{q}_1 \\ \vdots \\ \dot{q}_m \end{bmatrix}$$

$\underline{J}_{i/0}^L \in \mathbb{R}^{3 \times n}$ moving columns

$$\underline{w}_{i/0} = \sum_{j=1}^i \underline{J}_{i/j-1}^A \dot{q}_j \Rightarrow \underline{w}_{i/0} = \left[\underline{J}_{i/1}^A \quad \underline{J}_{i/2}^A \quad \dots \quad \underline{J}_{i/i}^A \right] \begin{bmatrix} \dot{q}_1 \\ \vdots \\ \dot{q}_m \end{bmatrix}$$

$\underline{J}_{i/0}^A \in \mathbb{R}^{3 \times n}$

$$\text{Then } \underline{J}_{i/0} = \begin{bmatrix} \underline{J}_{i/0}^A \\ \underline{J}_{i/0}^L \end{bmatrix} \in \mathbb{R}^{6 \times n}$$

If $i = n$ then

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$${}^0 J_{n/0} \equiv {}^0 J$$



base robot
Jacobian

NOTE Consider a point L fixed in frame i , then:

$$\begin{cases} \underline{v}_{L/0} = \underline{v}_{i/0} + \underline{\omega}_{i/0} \times \underline{z}_{L/i} \\ \underline{\omega}_{L/0} = \underline{\omega}_{i/0} \end{cases}$$

hence

$$\begin{bmatrix} {}^0 \underline{\omega}_{L/0} \\ {}^0 \underline{v}_{L/0} \end{bmatrix} = \underbrace{\begin{bmatrix} \underline{1} & 0 \\ -[\underline{z}_{L/i} \times] & \underline{1} \end{bmatrix}}_{\text{rigid body Jacobian}} \begin{bmatrix} {}^0 \underline{\omega}_{i/0} \\ {}^0 \underline{v}_{i/0} \end{bmatrix} = {}^0 S_{L/i} {}^0 J_{i/0} \begin{bmatrix} \dot{q}_1 \\ \vdots \\ \dot{q}_n \end{bmatrix}$$

rigid body Jacobian

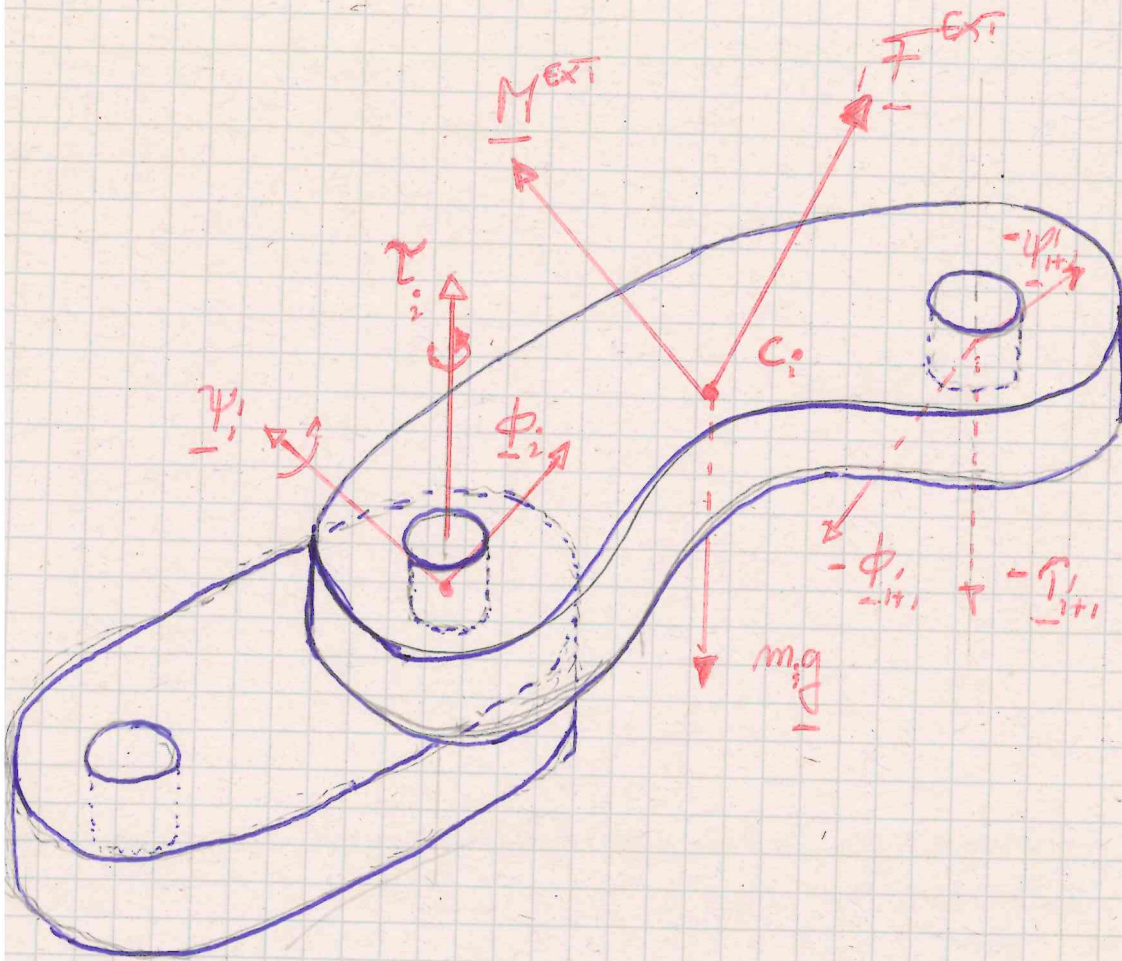
${}^0 S_{L/i}$

Then the end-effector velocity can be computed as

$$\begin{bmatrix} {}^0 \underline{\omega}_{E/0} \\ {}^0 \underline{v}_{E/0} \end{bmatrix} = {}^0 S_{E/0} {}^0 J_{n/0} \underline{\dot{q}}$$

FORCES/MOMENTS ACTING ON KINEMATIC CHAINS

8,



$\underline{F}^{EXT}, \underline{M}^{EXT}$

resultant of forces/moments external to the chain

$\underline{\Phi}_i, \underline{\Psi}_i$

reaction forces/torques due to joint i constraint

$-\underline{\Phi}_{i+1}, -\underline{\Psi}_{i+1}$

(counter) reaction forces/torques due to joint $i+1$

$\underline{T}_i$

actuation force (or torque) at joint i

$-\underline{T}_{i+1}$

(counter) actuation force (or torque) at joint $i+1$

VIRTUAL DISPLACEMENTS

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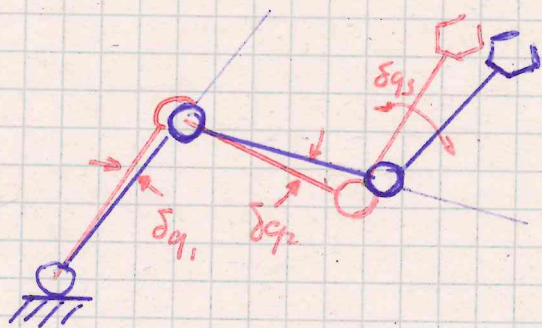
Considers an arbitrary infinitesimal displacement of the joint variables w.r.t. a given joint configuration and a given fixed time

We call virtual displacement of all the bodies of the system the infinitesimal change of positions and orientations (*)

$$\delta \underline{x}_i = \left(\frac{\partial \underline{x}_i}{\partial \underline{q}} \right) \delta \underline{q} = {}^0 \underline{J}_{x_i/o} \delta \underline{q}$$

$$\delta \underline{\theta}_i = \left(\frac{\partial \underline{\theta}_i}{\partial \underline{q}} \right) \delta \underline{q} = {}^0 \underline{J}_{\theta_i/o}^A \delta \underline{q}$$

symbolic
expression (see later)



*) The virtual displacements are not expressed as "differentials" to take into account the fact that time is a fixed assumed - For kinematic pairs (scleronic) " δ " $\equiv$ "d"